

RESEARCH DIRECTION · FROM AN EMPIRICAL RESULT TO A BET

Beyond Attention: what paradigm of intelligence can neuromorphic hardware actually ride?

If spiking hardware cannot ride the Transformer paradigm, what CAN it ride? **A map of six non-attention paradigms, and where I'd place the bet.**

Zeyu Wang · builds on “The Sparsity Ceiling” (arXiv, cs.NE)

What is the “point” of intelligence — and is attention it?

A working answer: intelligence is a **predictive generative model of the world, maintained in persistent dynamics, spending compute only on surprise, and acting to test its predictions.**

It is a **loop** (model $\rightleftharpoons$ world), not a **function** (context $\rightarrow$ token).

Transformers reach intelligence the opposite way: stateless, dense retrieval over the whole context, at scale.

THE EMPIRICAL HOOK

Our result — the sparsity ceiling:

Recurrence

hits a firing floor (~50%)

Attention

sparsifies, but $O(\text{context})$ KV = memory wall

Verdict

the Transformer route is hardware-mismatched to neuromorphic

They are opposites — and only one fits 20-watt hardware

DIMENSION	TRANSFORMER PARADIGM	BRAIN PARADIGM
State	stateless · re-reads full context	persistent recurrent dynamics
Memory	external KV cache (grows with context)	compressed into state + synapses
Compute spent on	dense all-pairs retrieval (QK ^T)	only prediction error (surprise)
Learning	offline · global backprop · huge data	online · local plasticity · continual
Source of intelligence	scale + retrieval	prediction + compression + action
Cost on neuromorphic	memory wall / dense MACs ✗	event-driven, analog, local ✓

The brain violates the assumption behind our own bound

Our firing-floor bound $\rho \geq H^{b-1}(\log_2 M / H)$ assumes information travels **only in spikes**. The brain breaks exactly that assumption — three ways:

Analog state, spikes for surprise

Memory lives in sub-threshold membrane + short-term synaptic state (activity-silent). Spikes signal only errors → escapes the firing floor AND the KV memory wall.

Enormous H

10^{11} neurons → $\log_2 M / H \approx 0 \rightarrow \rho_{\min} \rightarrow 0$. Sparse distributed coding is nearly free in high dimension.

Local, multi-timescale learning

Fast spikes / slow synapses / neuromodulation; local plasticity, continual — no backprop through all of history.

⇒ The brain is the **existence proof**: general intelligence at 20 W, ~1000× more efficient than GPUs — via prediction + sparsity + locality, not attention.

Six paradigms that reach intelligence without dense attention

PARADIGM	CORE MECHANISM	FIT	REPRESENTATIVE WORK (2024–26)
Predictive coding	hierarchical generative model; propagate only errors; local $\approx$ backprop	✓✓	Millidge+22; Salvatori/Song ICLR'24
Equilibrium propagation	relax to energy minimum; local weight updates; analog-native	✓✓	EqSpike; ImageNet-EP '26; -10^4 train energy
Analog / spiking SSM	linear recurrence, no KV cache, $O(L)$; state in analog dynamics	✓✓	Mamba; SpikingMamba; SpikySpace '26
World models / JEPA	predict in latent space (not tokens/pixels); plan by simulation	✓	LeCun V-JEPA 2; LeJEPA '25
Associative / Hopfield	content-addressable attractor retrieval = attention as dynamics	✓	Ramsauer; Hopfield–Fenchel–Young '24
Active inference	minimize expected surprise; perception–action loop; embodied	✓	Friston; scale-free active inference '25

✓✓ = neuromorphic physics is an advantage (analog, event-driven, local) · ✓ = algorithmic fit
BEYOND ATTENTION · NEUROMORPHIC INTELLIGENCE PARADIGMS

Where I would place neuromorphic intelligence

A predictive world-model held in analog recurrent dynamics,

learned locally & online (predictive coding / equilibrium propagation), acting in a closed loop — spikes carry only surprise.

Why this is not “settling” — the hardware–paradigm alignment:

■ Analog state escapes both traps

memory in membranes/synapses, not a KV cache → no memory wall; the firing floor is a spiking-state property (the bound itself: theory, scope limit tested)

■ Event-driven = compute only on surprise

predictive coding's error-only propagation IS event-driven — native to spiking silicon

■ Local plasticity = on-chip learning

EqProp/PC learn with local rules → training without global backprop, $\sim 10^4$ less energy

■ Recurrence + search = reasoning

test-time compute via iterated simulation/planning, not a giant single forward pass

The uncomfortable truth — and why it's still the right bet

NONE HAS BEATEN TRANSFORMERS

Predictive coding, EqProp, SSMs, Hopfield, JEPA, active inference — all promising, all more brain-like, **none has yet produced GPT-class general capability.**

This is a research bet, not a solved alternative. It should be stated as such.

BUT THERE IS AN EXISTENCE PROOF

The brain runs general intelligence on **20 watts**, ~1000× more efficient than GPUs — using exactly this paradigm: prediction, sparsity, locality, embodiment.

So the paradigm demonstrably CAN reach general intelligence. We just haven't reproduced it. That gap is the open frontier — and the reason to work on it.

Direction One, run: what the analog-state SSM actually did

01 Analog state buys energy, not quality

Param-matched 4-way SSM, char-LM, 3 seeds: analog state cuts proxy energy 1.60× (446k vs 712k pJ/token) for +0.160 bpc. “Beats digital” is retracted — the same noise on the digital baseline is worth −0.309 bpc.

02 The variant ranking inverts by workload

At matched communication rate the order inverts: char-LM analog $\ll$ spiking-state $\ll$ output-spiking; precise recall (copy) the reverse — margin kept 0.87 / 0.50 / 0.074. A 1-bit state is at exactly chance, and cheapest.

03 Degrade what the task ignores

One principle covers both: a degradation is cheap only where the loss does not depend on it. The same 6-bit state quantizer costs −0.002 bpc on char-LM and +0.37 on copy (post-hoc, not pre-registered).

04 My own bound did not survive its test

Shrink-H, 3 seeds per width: the predicted floor rises 12× while measured state activity falls 0.496 $\rightarrow$ 0.250, and accuracy improves as capacity shrinks. Theory with a tested scope limit, not a validated prediction.

05 The energy is not where the mechanisms are

The readout carries 75% of analog's pJ/token, the recurrence 8%; even a free recurrence costs 1.74×. Gating the readout recovers the energy but costs $\sim$ +1.0 bpc (char-LM) and all of copy's margin, 3 seeds. Converters: 0.00–0.33%.

What I would work on next

01 Analog state — done in simulation, open in silicon

Finished in simulation, including the energy datapath: the readout's 75% share resisted every gating mechanism tested. What is unfinished is physical — every pJ figure is a 45nm proxy. So 02 first, not more simulation.

02 Predictive coding on neuromorphic silicon

Implement an error-propagating hierarchy with local (EqProp) learning; measure real energy on Loihi 2 / SpiNNaker2 — turning the 45nm proxy into a measurement.

03 Quantify the floor-vs-wall trade-off

Formalize the recurrence(firing floor) ↔ attention(memory wall) dichotomy as a conservation law; find the Pareto frontier and where analog-state models sit on it.

04 Closed-loop world-model at the edge

A predictive world-model + planning loop for a low-power embodied agent — where neuromorphic's real-time, event-driven physics is decisive.

THESIS

Intelligence is a loop, not a lookup.

If neuromorphic can't ride attention, it shouldn't — its physics aligns with the **predictive, sparse, local, embodied** paradigm the brain already proves works at 20 W.

SOURCES Predictive coding: Millidge+22, Salvatori/Song ICLR'24 (2506.06332) · EqProp: Scellier, EqSpike, 2606.03584 · SSM: Mamba, SpikingMamba 2510.04595, SpikySpace · JEPA: V-JEPA 2, LeJEPA · Hopfield: Ramsauer, 2411.08590 · Active inference: Friston, PMC12217590 · Sparsity Ceiling (this work, arXiv cs.NE)